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Elabela as a predictive biomarker for placenta accreta spectrum: a case–control study

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Abstract

Objective To evaluate maternal serum Elabela (ELA) levels in pregnancies complicated by placenta accreta spectrum (PAS) and to assess their association with disease severity and predictive value for adverse obstetric outcomes.

Materials and methods In this prospective case–control study conducted at the Department of Perinatology, Ankara Etlik City Hospital (Türkiye) between February and July 2025, 88 pregnant women were included: 44 with PAS and 44 healthy controls matched for maternal age and gestational age. Maternal serum ELA levels were measured using a commercially available ELISA kit. Demographic, obstetric, clinical, and perinatal data were recorded. Receiver operating characteristic (ROC) curve analysis was performed to determine the diagnostic performance of ELA, and logistic regression was used to evaluate its predictive value for PAS.

Results Median ELA levels were significantly higher in the PAS group compared to controls (2600.38 [2419.52–2700.75] vs. 2359.65 [2249.25–2490.63] pg/mL; $p < 0.001$). The area under the ROC curve was 0.721 ($p < 0.001$; 95% CI: 0.615–0.811). The optimal cut-off value of 2488.75 pg/mL predicted PAS with 70.5% sensitivity, 75.0% specificity, a positive likelihood ratio of 2.82, and a negative likelihood ratio of 0.39.

Conclusion Elevated maternal serum ELA levels are associated with PAS and may serve as a potential biomarker for its prenatal identification. Further large-scale prospective studies are warranted to confirm its diagnostic utility and to explore its role in disease pathophysiology.

Keywords Placenta accreta spectrum, Elabela, Apelinergic system

Introduction

Placenta accreta spectrum (PAS) describes a group of placentation abnormalities in which the placenta attaches more deeply than normal to the uterine wall, sometimes penetrating into the serosa and even adjacent pelvic structures [1]. This serious obstetric condition greatly elevates the likelihood of severe maternal complications such as massive postpartum bleeding, emergency or planned hysterectomy, injuries to pelvic organs, the need for extensive blood transfusions, and other major surgical risks [2]. Detecting PAS before delivery, ideally during routine antenatal follow-up, is critical for reducing these risks [3, 4]. Management in a specialized, high-volume

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center by a multidisciplinary team—combined with well-timed delivery planning and thorough preoperative preparation—has been shown to help lower the rate of complications [5]. Two-dimensional ultrasonography is the most important diagnostic tool for this condition. Characteristic ultrasonographic features of PAS include loss of the clear uteroplacental interface, placental lacunae with turbulent flow, thinning or disruption of the myometrium, and increased vascularity extending into the bladder wall, all of which correlate with the severity of placental invasion [1, 6, 7]. Despite growing interest in serum biomarkers, there is currently no accurate, specific, and clinically validated blood test for antenatal diagnosis of PAS; evidence gaps include lack of standardized cut-offs, gestational-age effects, and external validation [8–10].

Elabela (ELA), also known as Apela, is a peptide hormone belonging to the apelinergic system, first identified in 2013–2014 [11]. Apelinergic signaling is critical for the functioning of angiogenesis, energy metabolism, fluid homeostasis and the cardiovascular system [11]. Recent studies have shown in such carcinomas that apelinergic signaling plays a crucial role in tumor neoangiogenesis [12]. Elabela plays a role in the renewal of embryonic stem cells in mice and in heart formation in zebrafish and has been detected in the kidneys and placenta of adult humans [13]. Later studies have demonstrated the expression of ELA in the human prostate, cardiac endothelium and blood vessels [14]. Interest in ELA as a marker of abnormal placentation has grown after a study by Demir et al., which found that maternal serum ELA concentrations were higher in women with placenta previa or accreta than in healthy pregnancies [15]. Although these findings suggest diagnostic promise, the study did not evaluate whether ELA levels relate to adverse maternal outcomes.

In light of these observations, we suggest that ELA could contribute to the pathophysiological processes involved in PAS. This study was therefore designed to explore both the diagnostic capacity and the clinical significance of maternal serum ELA levels in women with PAS. Alongside the comparison of ELA levels between cases and matched healthy controls, we examined relationships with clinically important outcomes such as intrapartum hemorrhage, need for hysterectomy, blood transfusion requirements, intensive care admission, and extended hospital stay.

Materials and methods

Study design and setting

This prospective case–control study was conducted between February and July 2025 in the Ankara Etlik City Hospital, a tertiary referral center in Ankara, Türkiye. The study was performed in accordance with the Declaration

of Helsinki and approved by the institutional ethics committee (Approval ID: AEŞH-BADEK-2025-0065). All participants provided written informed consent prior to inclusion. Ankara Etlik City Hospital has approximately 14,000 deliveries per year, with an overall cesarean section rate of about 58%.

Two-dimensional grayscale ultrasonography was used to assess the diagnosis of PAS. Well-documented sonographic findings examined during the study: Loss of clear zone, lacunae formation, increased vascularization in the placental bed, bridging of vessels through the bladder, and thinning of the myometrium under the placenta [1]. Patients diagnosed with PAS were hospitalized at 30–32 weeks of gestation for close monitoring and preoperative preparation. Betamethasone was administered to all patients in two doses, 24 h apart, at admission. The diagnosis of PAS was initially established antenatally based on characteristic ultrasonographic findings and subsequently confirmed intraoperatively and/or histopathologically according to the FIGO classification [1]. In our department, patients with placenta percreta are routinely managed by cesarean hysterectomy. In cases of placenta accreta or increta where hemorrhage can be controlled without hysterectomy, conservative techniques such as intrauterine balloon tamponade, compression sutures, or primary suturing of bleeding areas are applied. The control group consisted of an equal number of healthy pregnant women who were willing to participate in the study and were matched for maternal age, gestational age, body mass index (BMI), and gravida. Subjects were excluded from the study if they were unwilling or unable to provide written informed consent; had a history of chronic inflammatory disease, diabetes mellitus, or hypertension; were carrying multiple pregnancies; had clinical chorioamnionitis; or did not deliver within our institution. In addition, cases with placenta previa without invasion or isolated uterine dehiscence were excluded from the PAS group.

Data collection

Data were collected prospectively from pregnant women who were diagnosed with PAS and from healthy controls. The diagnosis of PAS was initially established antenatally based on characteristic ultrasonographic findings and subsequently confirmed intraoperatively and/or histopathologically according to the FIGO classification [1]. Histopathological confirmation was available only in patients who underwent hysterectomy (all cases of placenta percreta), whereas in cases managed conservatively (placenta accreta and increta), the diagnosis was based on antenatal imaging and intraoperative findings. Healthy controls were selected from women undergoing elective cesarean delivery without any ultrasound or intraoperative evidence of placental invasion. Controls

were matched with PAS cases for maternal age and gestational age at sampling. All participants were managed in the same tertiary center under identical perioperative protocols to minimize selection bias.

Maternal demographic characteristics, obstetric history (gravidity, parity, number and interval of previous cesarean sections, history of curettage, smoking status, and adverse obstetric history), and clinical parameters were recorded at the time of enrollment. Gestational age at blood sampling and estimated fetal weight were documented. Neonatal outcomes, including gestational age at delivery, birth weight, 1- and 5-minute Apgar scores, and neonatal complications (fetal distress, respiratory distress syndrome, neonatal intensive care unit admission, and neonatal death), were collected from medical records. Maternal outcomes, such as postpartum hemorrhage, estimated blood loss, transfusion requirements (erythrocyte suspension, fresh frozen plasma, fibrinogen concentrate), admission to the intensive care unit (ICU), necessity for hysterectomy, and composite adverse maternal outcomes (CAMO), were also recorded. PAS grading was based on intraoperative findings and confirmed histopathologically where available, according to the FIGO classification. CAMO was defined as the presence of one or more of the following: intrapartum hemorrhage exceeding 1500 mL, requirement for blood product transfusion, necessity for hysterectomy, ICU admission, or prolonged hospitalization. Prolonged hospitalization was defined as a length of hospital stay exceeding the median duration observed in the PAS group (4 days).

Serum Elabela measurement

Venous blood samples were collected from the antecubital vein of participants. After collection, the samples were allowed to clot at room temperature for 10–20 min, then centrifuged at 2000–3000 rpm for 20 min, as recommended by the manufacturer. The resulting serum was carefully separated and stored at -80°C until analysis. Maternal serum Elabela levels were measured using a commercially available enzyme-linked immunosorbent assay (ELISA) kit (Human Apelin receptor early endogenous ligand (Elabela) ELISA Kit, Shanghai Coon Koon Biotech Co., Ltd, Shanghai, China; Cat No: CK-bio-26548). The assay is based on double-antibody sandwich technology and provides a detection range of 20–2000 pg/mL, with a sensitivity of 10 pg/mL. According to the manufacturer, the intra-assay coefficient of variation (CV) was $<7\%$, and the inter-assay CV was $<10\%$. All samples were tested in duplicate to ensure reproducibility. The absorbance was measured at 450 nm using a calibrated microplate reader, and the final Elabela concentrations were calculated based on the standard curve and multiplied by a dilution factor of 5, as per kit

instructions. Results were reported in picograms per milliliter (pg/mL).

Sample size calculation

The sample size calculation was based on the effect size (Cohen's $d = 0.99$) reported in the study by Demir et al., titled “The role of maternal blood elabela levels in the prediction of placenta previa and accreta.” [15]. The calculation was performed using the G*Power software (version 3.1.9.7) for a two-tailed independent samples t-test. A significance level (α) of 0.05, a power ($1-\beta$) of 0.95, and an allocation ratio of 1 were used. Based on these parameters, a minimum of 22 participants per group (total $n = 44$) was determined to be sufficient to achieve adequate statistical power.

Statistical analysis

The Statistical Package for the Social Sciences Version 27.0 (IBM Corporation, Armonk, New York, US) was used to analyze the data. The Kolmogorov-Smirnov and Shapiro-Wilk test was used to determine the distribution of the data. For normally distributed continuous variables, the Independent Samples t-test was used, and the results were reported as mean $\pm$ standard deviation (SD). For non-normally distributed continuous variables, the Mann-Whitney U test was applied, and the results were expressed as median (interquartile range, IQR). Categorical variables were compared using the Chi-square test or Fisher's exact test, and the results were presented as frequency and percentage (n , %). The correlation between Elabela levels and clinical parameters was evaluated using Spearman's rank correlation for continuous variables, and point-biserial correlation analysis was used for Elabela levels and dichotomous variables. Receiver Operating Characteristic (ROC) curve analysis was performed to assess the diagnostic performance of Elabela in predicting PAS. The optimal cut-off value was determined based on the Youden index. Area under the curve (AUC), sensitivity, specificity, and positive/negative likelihood ratios ($+LR/-LR$) were calculated. To identify independent predictors of PAS, both univariable and multivariable binary logistic regression analyses were conducted. Odds ratios (OR), adjusted odds ratios (aOR), 95% confidence intervals (CI), and corresponding p -values were reported. The significance level was set at $p < 0.05$ for all analysis.

Results

A total of 88 pregnant women were included in the study, with 44 patients in the PAS group and 44 in the healthy control group. Demographic, obstetric, and clinical characteristics are summarized in Table 1. The number of previous cesarean sections ($p = 0.011$) and the duration since the last cesarean section ($p = 0.004$) were significantly higher in the PAS group compared to controls.

Table 1 Comparison of demographic, obstetric, and clinical parameters between PAS patients and healthy pregnant controls

Variables	PAS (n:44)	Control group (n:44)	p-value
Maternal age (years)	33 (28–36)	30 (29–33)	0.131 ^a
BMI (kg/m ²)	29.00 (26.70–33.72, 70.72)	27.34 (25.76–31.24)	0.174 ^a
Gravida (n)	3 (3–4)	3 (2–4)	0.371 ^a
Parity (n)	2 (1–2)	1 (1–2)	0.060 ^a
Abortus (n)	0 (0–1)	0 (0–1)	0.229 ^a
Dilatation and curettage (n)	0 (0–0)	0 (0–0)	< 0.99 ^a
Number of Previous cesarean sections (n)	1 (1–2)	1 (1–1)	0.011^a
Duration from previous C/S (years)*	3 (2–5)	2 (1–3)	0.004^a
Smoking (n,%)	3 (6.8%)	2 (4.5%)	> 0.99 ^a
Gestational age at blood sampling (weeks)	32.2 ± 1.7	32.1 ± 1.4	0.707 ^b
Gestational age at birth (weeks)	34.2 (33.5–35.3)	38.1 (36.4–39.2)	< 0.001^a
Duration of post-operative hospitalization (days)	4 (3–7)	2 (2–2)	< 0.001^a
ES transfusion (n,%)	24 (54.5%)	1 (2.3%)	< 0.001^c
FFP transfusion (n,%)	15 (34.1%)	-	< 0.001^c
Fibrinogen Concrete transfusion (n,%)	1 (2.3%)	-	> 0.99 ^c
Blood Product Transfusion requirement (n,%)	24 (54.1%)	1 (2.3%)	< 0.001 ^c
CAMO (n,%)	35 (79.5%)	6 (13.6%)	< 0.001^c
Elabela (pg/mL)	2600.38 (2419.52–2700.75, 52.75)	2359.65 (2249.25–2490.63, 25.63)	< 0.001^a

Abbreviations: PAS Placenta Accreta Spectrum, BMI Body Mass Index, C/S Cesarean Section, ES Erythrocyte Suspension, FFP Fresh Frozen Plasma, Tx Transfusion, CAMO Composite Adverse Maternal Outcome

^aNon-normally distributed parameter; analyzed using Mann–Whitney U test and presented as median (interquartile range)

^bNormally distributed parameter; analyzed using independent samples t-test and presented as mean ± SD

^cCategorical variables; analyzed using Chi-square test or Fisher's exact test as appropriate, and presented as number (percentage)

*"Duration from previous C/S (years)" was calculated only for women with at least one prior cesarean delivery. This variable was not applicable for 7 patients in the PAS group and 10 patients in the control group who were undergoing their first cesarean section

Gestational age at birth and length of postoperative hospitalization were significantly different between the groups ($p < 0.001$ for both). The PAS group had significantly higher rates of erythrocyte suspension transfusion, FFP transfusion, overall transfusion requirement, and CAMO ($p < 0.001$ for all). Median maternal serum Elabela levels were also significantly higher in the PAS group than in controls (2600.38 vs. 2359.65 pg/mL; $p < 0.001$).

Among PAS patients, comparisons of Elabela levels according to clinical outcomes and FIGO grades are shown in Table 2. There were no statistically significant differences in Elabela concentrations between CAMO

Table 2 Comparison of maternal serum Elabela levels in PAS cases according to clinical outcomes

Grouping Variable	Subgroup	Elabela, Median (IQR)	p-value
CAMO	CAMO (n = 35)	2598.50 (2417.65–2689.40, 65.40)	0.274 ^a
	Non-CAMO (n = 9)	2655.30 (2556.80–2727.25, 80.25)	
Hysterectomy	Required (n = 17)	2602.25 (2413.90–2674.25, 90.25)	0.470 ^a
	Not required (n = 27)	2598.50 (2421.40–2719.70, 40.70)	
Requirement of Blood Product Transfusion	Required (n = 24)	2649.63 (2505.73–2715.90, 73.90)	0.253 ^a
	Not required (n = 20)	2583.35 (2415.77–2679.93, 77.93)	
PAS Grade according to FIGO classification	Grade 1 (n = 13)	2643.95 (2316.60–2746.20, 60.20)	0.693 ^b
	Grade 2 (n = 14)	2585.23 (2518.95–2689.40, 95.40)	
	Grade 3a (n = 9)	2443.85 (2395.20–2602.25, 20.25)	
	Grade 3b (n = 6)	2655.30 (2590.90–2700.75, 90.75)	
	Grade 3c (n = 2)	2649.63 (2625.00–2674.25, 00.25)	

Abbreviations: PAS Placenta Accreta Spectrum, CAMO Composite Adverse Maternal Outcome

^aNon-normally distributed parameter; analyzed using Mann–Whitney U test and presented as median (interquartile range)

^bNon-normally distributed parameter; analyzed using Kruskal Wallis test and presented as median (interquartile range)

and non-CAMO patients ($p = 0.274$), between those requiring and not requiring hysterectomy ($p = 0.470$), or between those with and without blood product transfusion ($p = 0.253$). Among the patients with PAS, 13 (29.5%) were classified as FIGO grade 1, 14 (31.8%) as grade 2, 9 (20.4%) as grade 3a, 6 (13.6%) as grade 3b, and 2 (4.5%) as grade 3c. There were no statistically significant differences in Elabela levels among the groups according to FIGO grade ($p = 0.693$).

Correlation analysis results are presented in Table 3. In the overall population, Elabela levels were positively correlated with the presence of PAS ($r = 0.382$, $p < 0.001$). In the control group, no significant correlation was found between Elabela levels and maternal age, gestational age at blood sampling, BMI, or estimated fetal weight (all $p > 0.05$).

The diagnostic performance of Elabela in predicting PAS is presented in Table 4; Figure 1. The area under the ROC curve (AUC) was 0.721 (95% CI: 0.615–0.811; $p < 0.001$). The optimal cut-off value for Elabela was > 2488.75 pg/mL, yielding a sensitivity of 70.5% and a specificity of 75.0%. The positive likelihood ratio was 2.82, and the negative likelihood ratio was 0.39.

Table 3 Correlation between Elabela levels and clinical parameters in PAS and control groups

Variables	<i>r</i>	<i>p</i> -value
Overall population		
PAS	0.382	< 0.001 ^a
In the Control Group		
Maternal Age (years)	0.073	0.636 ^b
GA at Blood Sampling (weeks)	0.230	0.133 ^b
BMI (kg/m ²)	0.117	0.451 ^b
Estimated Fetal Weight (g)	0.071	0.647 ^b

^bSpearman's rank correlation was used for the analysis of non-normally distributed continuous variables

^aThe correlation between a continuous variable and dichotomous variables was assessed using point-biserial correlation analysis

Univariable and multivariable logistic regression analyses are shown in Table 5. Due to 7 patients in the PAS group and 10 in the control group were undergoing their first cesarean section, the variable “duration since the previous cesarean section” was not applicable to all participants and was therefore excluded from both the univariable and multivariable models to avoid bias. In the univariable analysis, the number of previous cesarean sections (OR: 2.050, 95% CI: 1.227–3.425; $p = 0.006$) and Elabela level (OR: 1.002, 95% CI: 1.001–1.004; $p = 0.009$) were significantly associated with PAS. In the multivariable model, the number of previous cesarean sections (aOR: 2.530, 95% CI: 1.392–4.601; $p = 0.002$) and Elabela

Table 4 Diagnostic performance of Elabela in predicting placenta accreta spectrum (PAS) based on ROC analysis

Variables	Cut-off	AUC	95% CI	Sensitivity (%)	Specificity (%)	+LR	-LR	<i>p</i> -value
Elabela (pg/mL)	> 2488.75	0.721	0.615–0.811	70.5	75.0	2.82	0.39	< 0.001

Abbreviations: ROC Receiver Operating Characteristic, AUC Area Under the Curve, CI Confidence Interval, +LR Positive Likelihood Ratio, -LR Negative Likelihood Ratio

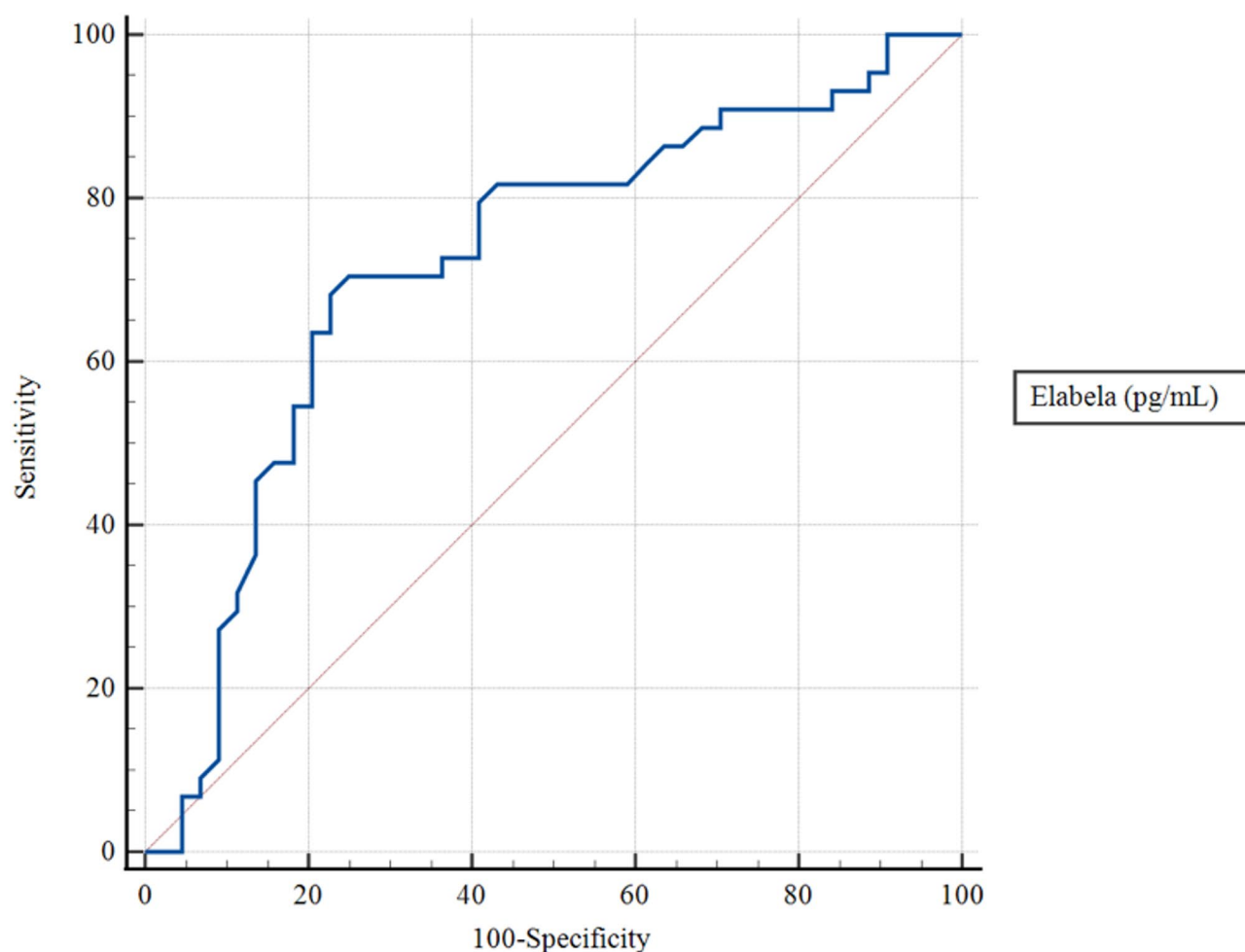
**Fig. 1** Receiver Operating Characteristic (ROC) curve showing the diagnostic performance of maternal serum Elabela levels for predicting placenta accreta spectrum (PAS)

Table 5 Univariable and multivariable logistic regression analyses for prediction of placenta accreta spectrum (PAS)

Variables	Univariable			Multivariable		
	OR	95% CI	p-value	aOR	95% CI	p-value
Maternal Age (years)	1.105	0.984–1.242	0.093	1.114	0.978–1.270	0.104
Number of Previous C/S	2.050	1.227–3.425	0.006	2.530	1.392–4.601	0.002
Elabela	1.002	1.001–1.004	0.009	1.003	1.001–1.005	0.003

Abbreviations: OR Odds Ratio, aOR Adjusted Odds Ratio, CI Confidence Interval

remained a statistically significant predictor of PAS (aOR: 1.003, 95% CI: 1.001–1.005; $p = 0.003$).

Discussion

In this prospective case–control study, maternal serum Elabela (ELA) concentrations were found to be higher in pregnancies complicated by placenta accreta spectrum (PAS) compared with gestational age–matched healthy controls. The area under the curve (AUC) value obtained from the ROC analysis was 0.721, corresponding to a sensitivity of 70.5% and a specificity of 75.0%, indicating that ELA has moderate diagnostic power in distinguishing PAS. However, ELA levels did not show a significant association with combined adverse maternal outcomes (CAMO), which include intrapartum hemorrhage, transfusion requirement, hysterectomy, admission to the intensive care unit (ICU), or prolonged hospital stay. Additionally, no differences were found in ELA levels according to PAS grades. These findings suggest that while ELA may contribute to the pathophysiology of PAS, its role as a marker for clinical severity or adverse outcomes is limited.

ELA is a peptide that is an endogenous ligand of the apelin receptor and an important component of the apelinergic system. This system plays a role in fundamental physiological processes such as angiogenesis, cardiovascular function, embryonic development, energy metabolism, and fluid balance [11]. ELA, which is highly expressed in trophoblast cells in the placenta, has been shown to promote extravillous trophoblast invasion, modulate placental hormone secretion, and maintain vascular integrity throughout pregnancy [14]. These properties suggest that ELA may play a role in the pathogenesis of abnormal placentation. The possible association between Elabela and PAS can be explained by its physiological role in placental development. Elabela, an endogenous ligand of the apelin receptor (APJ), is expressed in trophoblastic tissue and contributes to angiogenesis, vascular remodeling, and trophoblast invasion, which are all essential for normal placentation [14, 16]. Disturbances in these mechanisms may result in excessive trophoblastic invasion and defective decidualization, both of which play a central role in the development of PAS.

Several studies have examined maternal serum ELA levels in obstetric complications. Huang et al. observed

that ELA levels were elevated in hypertensive pregnancies during the first and second trimesters, but not in the third, while no significant difference was observed in preeclampsia, suggesting gestational age-dependent variability [17]. Panaitescu et al. reported higher ELA concentrations in late-onset preeclampsia compared to early-onset disease and normotensive controls [18]. Additionally, Pécheux et al. highlighted the role of ELA and the apelinergic system in placental proliferation, invasion, apoptosis, and hormone production. They proposed that dysregulation of this system contributes to placental pathologies such as preeclampsia, fetal growth restriction, and gestational diabetes mellitus [14]. Taken together, these findings indicate that ELA behaves as a context- and time-sensitive marker of placental adaptation rather than a stand-alone diagnostic indicator.

The data regarding ELA in PAS remain limited. Demir et al. examined maternal ELA levels in a heterogeneous group of patients, including cases of placenta previa and PAS, and found significantly higher ELA concentrations in both groups compared to the control group [15]. The researchers suggested that ELA may be helpful in distinguishing abnormal types of placental attachment. However, their study had some limitations, including a small sample size, the absence of a control group matched for gestational age, the lack of analysis of adverse maternal outcomes, and limited diagnostic performance in PAS diagnosis (AUC: 0.658). Our study, on the other hand, focused solely on PAS cases, used a larger and gestational age-matched cohort, and provided both ultrasonographic and histopathological confirmation. Additionally, clinically meaningful outcomes such as CAMO were analysed, and the relationship between ELA levels and PAS grades was also evaluated.

In addition to ELA, various biomarkers have been investigated for use in the diagnosis of PAS. Zhang et al. [19] reported that low placental growth factor (PlGF) and high soluble fms-like tyrosine kinase-1 (sFlt-1) levels may be beneficial in diagnosing PAS, but noted that their diagnostic performance was limited and that their AUC values were comparable to those we found for ELA. Similarly, Cai et al. [20] observed elevated maternal serum alpha-fetoprotein (AFP) levels in PAS pregnancies during the second trimester, but noted that specificity was inadequate. Gong et al. [21] proposed coagulation parameters such as elevated D-dimer and decreased

fibrinogen levels as potential markers; however, these reflect systemic haemostatic changes rather than placental biology. In contrast, ELA plays a direct role in angiogenesis and trophoblast invasion, resembling the vascular endothelial growth factor (VEGF) pathway in this regard. Farisoğulları et al. reported that VEGF levels were elevated in both PAS and placenta previa, reflecting increased angiogenic activity but showing limited discriminatory power between the two conditions [22]. The combined use of angiogenic markers such as ELA and VEGF may therefore enhance diagnostic accuracy.

The interpretation of diagnostic performance should be contextualized within the clinical setting. With an AUC of 0.721 indicating moderate discriminative ability, and $+LR \approx 2.8$ and $-LR \approx 0.39$ values, ELA alone is limited as a “screening test”; however, when added to a multi-parameter algorithm in patients with high pre-test probability, it may be useful. These findings are consistent with an approach that complements imaging findings (e.g., clear zone loss, lacunar formation, increased vascularity in the placental bed, bladder bridging vessels), which remain the gold standard in the antenatal diagnosis of PAS, with a complementary biomarker. Although no significant association was found between ELA levels and clinical outcomes such as CAMO or PAS grade, these analyses were exploratory and may have been underpowered due to limited subgroup sizes. Therefore, ELA should presently be interpreted as a diagnostic rather than prognostic marker until validated in larger, outcome-oriented cohorts.

In the multivariable logistic regression analysis, maternal serum Elabela levels remained significantly associated with the presence of PAS, together with the number of previous cesarean sections. This finding suggests that ELA may have a role in the mechanisms leading to abnormal placental invasion and could assist in identifying women at increased risk. Nevertheless, it is not intended to replace imaging-based diagnosis. Considering that ultrasound is the main tool for detecting PAS, ELA should be evaluated as a possible supporting biomarker alongside clinical risk factors and imaging findings within an integrated risk assessment approach.

One of the most important strengths of this study is its prospective design and the use of a control group matched according to gestational age. In addition, the exclusion of comorbidities that could affect biomarker levels minimised the impact of potential confounding factors. The confirmation of PAS diagnosis with prenatal ultrasound findings and intraoperative/histopathological verification has reduced the risk of misclassification. Focusing not only on diagnostic features but also evaluating clinical outcomes such as CAMO and other complications provides additional information on the potential clinical value of ELA.

However, some limitations should be considered. The fact that the study was conducted at a single tertiary centre may limit the generalisability of the results to different clinical settings or broader populations. The fact that ELA levels were measured only once in the third trimester prevented the identification of possible trends in changes throughout pregnancy. Longitudinal measurements covering all trimesters could allow for a more comprehensive evaluation of diagnostic potential. Although the sample size was statistically sufficient, it may be limited in terms of detecting small effects, especially in subgroup analyses. Finally, although ELA levels were associated with PAS severity and adverse outcomes, other placental biomarkers or histopathological variables were not included in the study. These additional data could have contributed to a more profound understanding of the biological mechanisms involved.

The data obtained indicate that ELA is associated with the presence of PAS but cannot predict clinical severity or adverse outcomes on its own. Therefore, it appears more appropriate to evaluate ELA as a complementary marker alongside prenatal ultrasound and risk factor profiling (e.g., number and interval of previous cesarean sections). Multicentre, larger-sample, longitudinal studies covering all trimesters, along with comparative analyses of multi-biomarker panels, will provide a clearer understanding of ELA's time-sensitive diagnostic value and its additional contribution to clinical decision-making processes.

Conclusion

In this study, higher maternal serum ELA levels in PAS pregnancies support the possible role of this peptide in the pathophysiology of abnormal placentation. Although its diagnostic power is limited on its own, the inclusion of ELA in integrated risk models alongside imaging, clinical history, and other placental biomarkers may improve early risk classification and surgical planning, particularly in high-risk pregnancies. Further studies with large sample sizes, multiple centres, and coverage of all trimesters are needed to clarify the timing of measurement, determine clinically meaningful threshold values, and evaluate its additional contribution to patient outcomes.

Abbreviations

AFP	Alpha-Fetoprotein
AUC	Area Under the Curve
BMI	Body Mass Index
CAMO	Composite Adverse Maternal Outcome
CI	Confidence Interval
C/S	Cesarean Section
CV	Coefficient of Variation
ELA	Elabela
ELISA	Enzyme-Linked Immunosorbent Assay
ES	Erythrocyte Suspension
FFP	Fresh Frozen Plasma
FIGO	International Federation of Gynecology and Obstetrics

GA	Gestational Age
ICU	Intensive Care Unit
IQR	Interquartile Range
LR	Likelihood Ratio
OR	Odds Ratio
PAS	Placenta Accreta Spectrum
PLGF	Placental Growth Factor
ROC	Receiver Operating Characteristic
sFlt-1	Soluble fms-like tyrosine kinase-1
SD	Standard Deviation
Tx	Transfusion
VEGF	Vascular Endothelial Growth Factor

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Authors' contributions

ÖVA: Project administration, Writing – original draft. DS: Writing – original draft, Conceptualization. RTA: Writing – review & editing, Methodology. AKu: Writing – review & editing, Validation. AKi: Data curation, Resources. AAF: Formal analysis, Data curation. BSÜ: Data curation, Investigation. ATÇ: Supervision, Project development. KYY: Supervision, Project administration.

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Data availability

Due to hospital policies, patient data and study materials cannot be shared. However, the data are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate

All patients gave informed consent to be enrolled in this project and the study was approved by the local ethics committee of the institution (No. AEŞH-BADEK-2025-0065) and was conducted in accordance with the Declaration of Helsinki. The data were anonymized and handled in compliance with ethical guidelines to ensure participant confidentiality.

Consent for publication

Not applicable. This study does not contain any individual person's data in any form (including any individual details, images, or videos) that would require consent for publication.

Competing interests

The authors declare no competing interests.

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